

Forder — Book IV

Proposition 16

Formal Reconstruction — Technical Documentation

Jurek Rózański
Military University of Technology, Warsaw, Poland
`jerzy.rozanski@wat.edu.pl`

CGJTEAMLAB

If A, B, C, D lie on one circle, then the angles subtended by AB at C, D are supplementary when C, D lie on opposite sides of AB , and congruent when they lie on the same side.

1 Introduction

Forder IV.16 collects the two fundamental “same chord” configurations for four points on one circle.

Let A, B, C, D lie on a circle. The chord AB separates the plane into two half-planes. There are two geometrically different possibilities.

If C, D lie on opposite sides of AB , then the two angles

$$\angle ACB, \quad \angle ADB$$

are supplementary.

If C, D lie on the same side of AB , then

$$\angle ACB \cong \angle ADB.$$

Forder states both conclusions in one proposition [1]. The production library deliberately exposes them as two separate proposition interfaces and two separate public theorems:

```
hilbert_IV16_opposite_side  
hilbert_IV16_same_side
```

This separation is mathematically useful. Later proofs—especially IV.17, IV.19, and IV.20—need to invoke one side configuration at a time. The production API therefore preserves the actual branching structure rather than hiding it inside a disjunction.

The implementation lives in

2 Forder's Statement

Forder IV.16 says:

If A, B, C, D lie on a circle and C, D are on opposite sides of AB , then the angles ACB, ADB are supplementary; if C, D are on the same side of AB , then these angles are congruent.

His proof treats the two halves differently.

The opposite-side half follows immediately from IV.15 by viewing C, A, D, B as a cyclic quadrilateral. The opposite angles of that quadrilateral are precisely

$$\angle ACB \quad \text{and} \quad \angle ADB.$$

For the same-side half, Forder distinguishes the position of the centre relative to chord AB .

If the centre lies on AB , then AB is a diameter and both inscribed angles are right by IV.13.

If C, D and the centre lie in the same half-plane, then the result is exactly IV.12.1.

If C, D lie on one side of AB and the centre lies on the other, an antipodal point X is introduced. Both

$$\angle ACB \quad \text{and} \quad \angle ADB$$

become supplements of the same angle

$$\angle AXB.$$

Hence they are congruent.

That three-case source argument is preserved explicitly in the formal proof.

3 Formal Interfaces

The two proposition contracts are defined in `HilbertInterfaceIV.lean`.

3.1 Same-side interface

```
def HilbertForderIV16SameSide
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall K R A B C E : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
        H.OnLine A chord ->
          H.OnLine B chord ->
            HilbertCircle Geo K R A ->
```

```

HilbertCircle Geo K R B ->
HilbertCircle Geo K R C ->
HilbertCircle Geo K R E ->
HilbertSameSide Geo C E chord ->
Geo.AngleCongruent
  A C B
  A E B

```

The conclusion is direct angle congruence.

3.2 Opposite-side interface

```

def HilbertForderIV16OppositeSide
  [H : HilbertIncidence Geo]
  [_HO : @HilbertOrder Geo H] : Prop :=
  forall K R A B C E : Geo.Point,
    forall chord : Geo.Line,
      Ne A B ->
      H.OnLine A chord ->
      H.OnLine B chord ->
      HilbertCircle Geo K R A ->
      HilbertCircle Geo K R B ->
      HilbertCircle Geo K R C ->
      HilbertCircle Geo K R E ->
      HilbertOppositeSide Geo C E chord ->
      exists X : Geo.Point,
        BookZeroSupplement Geo
          A E B
          B X /\
          Geo.AngleCongruent
            A C B
            X E B

```

The supplementary relation is therefore constructive. The witness X extends AE through E , and the theorem says that the adjacent angle

$$\angle XEB$$

is congruent to

$$\angle ACB.$$

The production endpoints are:

```

theorem hilbert_IV16_opposite_side
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV16OppositeSide (Geo := Geo)

theorem hilbert_IV16_same_side
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV16SameSide (Geo := Geo)

```

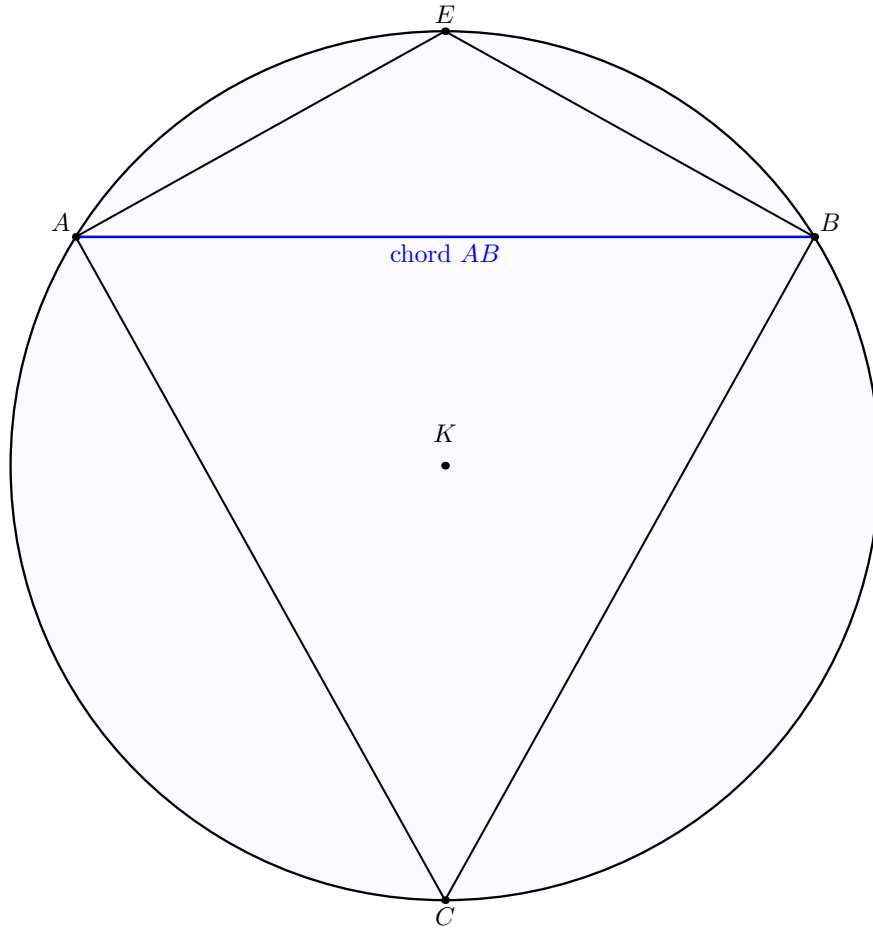
4 Opposite-Side Half

The opposite-side half is not an independent new circle theorem. It is an immediate reformulation of IV.15.

Suppose C, E lie on opposite sides of chord AB . Consider the cyclic quadrilateral

$$C - A - E - B.$$

apply IV.15 to $C - A - E - B$



C, E opposite sides of AB

Figure 1: Opposite-side half of IV.16. The four circle points are viewed in the cyclic order $C - A - E - B$. IV.15 applied to this quadrilateral says that the opposite angles $\angle ACB$ and $\angle AEB$ are supplementary.

Its diagonal is still the original chord AB . The two opposite vertices are C, E , and by assumption they lie on opposite sides of that diagonal.

The two opposite angles in IV.15 become:

$$\angle ACB$$

at C , and

$$\angle AEB$$

at E .

IV.15 therefore constructs X with

$$A - E - X$$

and

$$\angle ACB \cong \angle XEB.$$

This is exactly the definition of `HilbertForderIV16OppositeSide`.

The reduction is encapsulated by the private production helper

```
hilbert_forder_IV16_opposite_of_IV15_stage13
```

and the closed public theorem simply supplies IV.15:

```
theorem hilbert_IV16_opposite_side
  [H : HilbertIncidence Geo]
  [HE : @HilbertEuclideanPlane Geo H] :
  HilbertForderIV16OppositeSide (Geo := Geo)
```

This branch is therefore conceptually one line:

$$\boxed{\text{IV.16 opposite-side} = \text{IV.15 on } C, A, E, B.}$$

5 Same-Side Half: Why Three Cases Are Needed

Now assume that C, D lie on the same side of chord AB .

The conclusion is

$$\angle ACB \cong \angle ADB.$$

The source proof divides according to the position of the centre K relative to the chord AB . The formal proof follows exactly that classification.

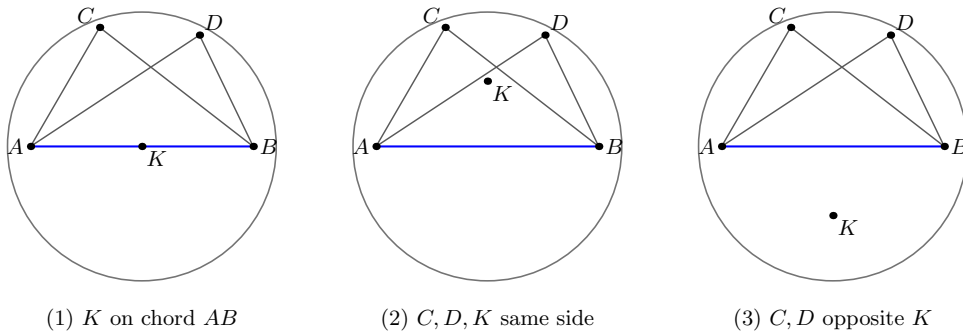


Figure 2: The three geometric cases in the same-side half of IV.16: (1) the centre K lies on AB ; (2) C, D, K occupy one half-plane; (3) C, D occupy one half-plane while K lies in the other.

The three cases are:

1. K lies on AB ;

2. K is off AB and lies on the same side as C, D ;
3. K is off AB and lies on the opposite side from C, D .

The first is the diameter case, the second is IV.12.1 directly, and the third is the only genuinely new reduction.

6 Case I: the Centre Lies on the Chord

Assume

$$K \in AB.$$

Then AB is a diameter.

IV.13 applied at C gives

$$\angle ACB \text{ right.}$$

IV.13 applied at D gives

$$\angle ADB \text{ right.}$$

The theorem

```
hilbert_all_right_angles_congruent
```

then yields

$$\angle ACB \cong \angle ADB.$$

No same-segment theorem is required in this branch.

This case is a useful reminder that the same-side theorem must include the possibility that the centre lies on the chord carrier. The relation

$$“C \text{ and } K \text{ are on the same side of } AB”$$

is not even defined when K lies on the line, so the diameter case cannot be subsumed under IV.12.1.

7 Case II: the Centre Lies in the Same Half-Plane

Assume now that K is off chord AB and

$$C \sim K$$

with respect to the chord carrier.

Since C, D are on the same side of AB , symmetry and transitivity of same-side relations give

$$D \sim K.$$

All three points C, D, K therefore lie in the same half-plane determined by AB .

This is precisely the hypothesis of IV.12.1. Hence

$$\boxed{\angle ACB \cong \angle ADB}.$$

The formal proof in this branch is correspondingly short:

```

have hSameDC :
  HilbertSameSide Geo D C chord := ...

have hSameDK :
  HilbertSameSide Geo D K chord := ...

exact
  hilbert_IV12_same_segment
    Geo K R A B C D chord ...

```

Thus the ordinary “same segment” theorem is recovered as the middle branch of the more general IV.16 same-side statement.

8 Case III: the Centre Lies on the Opposite Side

The remaining case is the important one.

Assume:

$$C \sim D$$

relative to chord AB , but

$$C \not\sim K.$$

Since both C and K are off the chord, they are on opposite sides:

$$C \mid K.$$

Because D is on the same side as C , it follows that

$$D \mid K.$$

Now construct the antipode X of C :

$$C - K - X.$$

The next task is to prove

$$C \mid X$$

relative to AB .

The opposite-side witness for C, K supplies a point Y with

$$C - Y - K$$

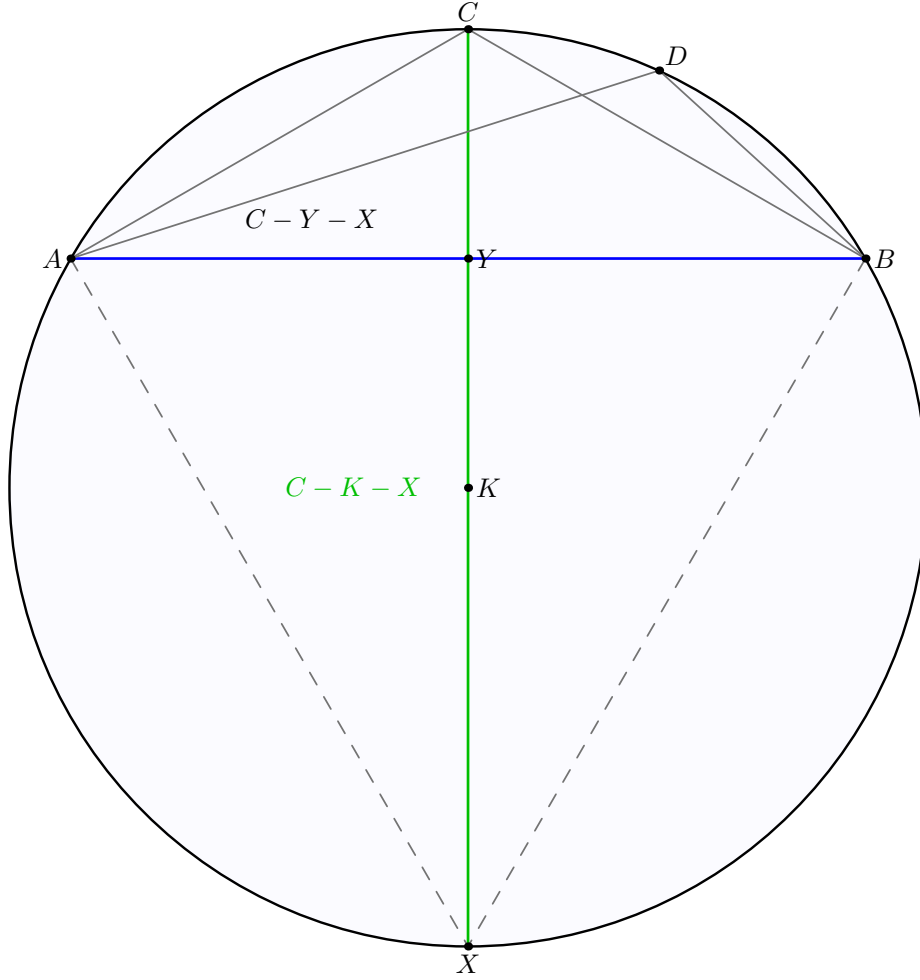
on chord AB . Since

$$C - K - X,$$

inner betweenness transitivity gives

$$C - Y - X.$$

C, D same side of AB



X opposite both C and D

Figure 3: Third same-side case. The centre K lies on the opposite side of chord AB from C, D . The antipode X of C satisfies $C - K - X$ and lies on the K -side. Therefore both pairs C, X and D, X are opposite-side configurations for chord AB .

Thus the same point Y witnesses that C and X are on opposite sides of the chord.

Because $D \sim C$, transport of opposite-side relations gives

$$D \mid X.$$

At this point the already proved opposite-side half of IV.16 can be applied twice.

9 Two Supplements of the Same Angle

Apply the opposite-side theorem to C, X . It produces a supplement of

$$\angle AXB$$

congruent to

$$\angle ACB.$$

Concretely there exists U such that

$$\text{Supplement}(AXB, BXU)$$

and

$$\angle ACB \cong \angle UXB.$$

Apply the opposite-side theorem again to D, X . It produces V such that

$$\text{Supplement}(AXB, BXV)$$

and

$$\angle ADB \cong \angle VXB.$$

Therefore $\angle UXB$ and $\angle VXB$ are two supplements of the same proper angle $\angle AXB$.

Book Zero theorem 43 on supplements gives

$$\angle UXB \cong \angle VXB.$$

Transitivity now yields

$$\angle ACB \cong \angle UXB \cong \angle VXB \cong \angle ADB.$$

Hence

$$\boxed{\angle ACB \cong \angle ADB}.$$

This is Forder's source idea expressed without numerical angle arithmetic: two angles supplementary to the same angle are congruent.

10 Why the Antipode Is the Correct Auxiliary Point

The point X is not chosen merely because it lies conveniently on the circle.

The antipodal order

$$C - K - X$$

has two essential consequences.

First, it places X on the opposite side of chord AB from C when the centre is already opposite C .

Second, because D lies on the same side as C , the same X is also opposite D .

Thus a single auxiliary circle point creates two opposite-side IV.16 configurations:

$$(C, X) \quad \text{and} \quad (D, X).$$

The resulting supplements share the same reference angle

$$\angle AXB.$$

That common reference is exactly what allows Book Zero supplement uniqueness to close the proof.

11 Lean Formalization

The production file contains a small public surface:

```
hilbert_IV16_opposite_side
hilbert_IV16_same_side
```

The proposition-specific reduction layer is represented by private helpers whose historical stage suffixes are not part of the public API:

```
hilbert_forder_IV16_opposite_of_IV15_stage13
hilbert_forder_IV16_same_of_IV12_1_IV13_IV15_stage15
hilbert_forder_IV16_opposite_complete_stage53
hilbert_forder_IV16_same_complete_stage54
```

Conceptually the closed wrappers amount to

```
theorem hilbert_IV16_opposite_side ... :=
  hilbert_forder_IV16_opposite_of_IV15_stage13
  Geo
  (hilbert_IV15_cyclic_supplement Geo)

theorem hilbert_IV16_same_side ... :=
  hilbert_forder_IV16_same_of_IV12_1_IV13_IV15_stage15
  Geo
  (hilbert_IV12_same_segment Geo)
  (hilbert_IV13_diameter_right_angle Geo)
```

```
(hilbert_IV15_cyclic_supplement Geo)
```

The exact internal production implementation may inline or wrap these private declarations, but the mathematical dependency is the one displayed above.

12 Production Dependency Structure

The two halves have different direct logical dependencies.

The opposite-side half is:

$$\text{IV.15} \implies \text{IV.16 opposite-side.}$$

The same-side half is:

$$\text{IV.12.1} + \text{IV.13} + \text{IV.15} \implies \text{IV.16 same-side.}$$

At module level the production file imports:

```
HilbertInterfaceIV
Proposition4_12
Proposition4_13
Proposition4_15
  -> Proposition4_16
```

There is no direct IV.14 import because IV.15 already encapsulates the supplement theorem needed downstream.

13 Relation to the Previous Propositions

13.1 IV.12.1

IV.12.1 solves exactly the same-side case in which the centre lies in the same segment as C, D .

13.2 IV.13

IV.13 solves the exceptional case in which chord AB passes through the centre and is therefore a diameter.

13.3 IV.15

IV.15 has two roles.

First, it immediately gives the opposite-side half of IV.16.

Second, through that opposite-side half, it is used twice in the antipodal case of the same-side proof.

Thus IV.16 is not merely “a restatement of IV.15”. One half is a direct corollary, while the other combines the earlier same-segment theorem, diameter theorem, and supplement theorem according to the position of the centre.

14 Hilbert and Book Zero Components

14.1 Plane separation

Same-side and opposite-side relations are the organizing language of the entire proposition. The case split is fundamentally a classification of the centre relative to the chord.

The formal proof uses:

- symmetry and transitivity of same-side;
- conversion of “not same side” into opposite-side when both points are off the carrier;
- transport of opposite-side relations across a same-side pair.

14.2 Betweenness

The antipodal relation

$$C - K - X$$

is combined with the crossing witness

$$C - Y - K$$

to obtain

$$C - Y - X.$$

This is the exact order step that turns the antipodal construction into an opposite-side statement.

14.3 Congruence of right angles

In the diameter branch both target angles are right by IV.13. The proof uses the general theorem that all right angles are congruent.

14.4 Book Zero supplements

The difficult same-side branch closes because two constructed angles are supplements of the same proper angle AXB . Book Zero theorem 43 turns that shared supplement relation into angle congruence.

No numerical representation of a supplement is used.

15 Foundational Boundary

IV.16 introduces no new axiom and no new circle-existence principle.

Its content is assembled from:

- the existing circle predicate;
- plane separation and betweenness;
- the antipode construction;
- IV.12.1;
- IV.13;
- IV.15;
- Book Zero supplement transport.

The theorem therefore shows especially clearly how a later classical result can be built by recombining earlier synthetic layers without adding new foundational commitments.

16 Classical and Formal Dependency Diagrams

The opposite-side half is minimal:

$$\begin{array}{c}
 C \mid D \text{ across chord } AB \\
 \downarrow \\
 C - A - D - B \text{ cyclic quadrilateral} \\
 \downarrow \\
 \text{IV.15} \\
 \downarrow \\
 \angle ACB \text{ supplementary to } \angle ADB.
 \end{array}$$

The same-side half is:

$$\begin{array}{c}
 C \sim D \text{ across } AB \\
 \downarrow \\
 \text{position of } K \\
 \downarrow \\
 \left\{ \begin{array}{ll}
 K \in AB & \xRightarrow{\text{IV.13}} \text{both angles right,} \\
 C \sim K & \xRightarrow{\text{IV.12.1}} \angle ACB \cong \angle ADB, \\
 C \mid K & \implies C - K - X \implies \begin{cases} C \mid X, \\ D \mid X \end{cases} \\
 & \implies \text{two IV.16 opposite-side supplements of } \angle AXB \\
 & \implies \angle ACB \cong \angle ADB.
 \end{array} \right.
 \end{array}$$

The module dependency graph is:

$$\begin{array}{ccccc}
 \text{IV.12} & \text{IV.13} & \text{IV.15} & & \\
 & \searrow & \downarrow & \swarrow & \\
 & & \text{IV.16} & &
 \end{array}$$

17 Diagrammatic Audit

Three figures are sufficient.

1. The opposite-side figure shows the direct IV.15 relabelling.
2. The three-case figure records the complete classification of the centre for the same-side half.
3. The antipode figure explains the only nontrivial same-side branch and makes visible why the same point X creates two opposite-side configurations.

No separate figure is needed for the diameter case; it is already a standard IV.13 configuration.

18 Conclusion

Forder IV.16 contains two theorems with quite different proof structures.

The opposite-side half is a direct corollary of IV.15 after relabelling the cyclic quadrilateral.

The same-side half requires a complete classification of the centre. If the centre lies on the chord, IV.13 gives two right angles. If it lies in the same half-plane, IV.12.1 applies directly. If it lies in the opposite half-plane, an antipodal point converts the problem into two applications of the opposite-side theorem, and Book Zero supplement uniqueness identifies the two target angles.

The production API reflects this structure by exporting separate same-side and opposite-side theorems. That choice is not merely organizational: it provides the exact interfaces required by the tangent and converse theorems that follow.

References

- [1] Henry George Forder, *Foundations of Euclidean Geometry*, Cambridge University Press, Cambridge, 1927.